Mini-Project 1: Image Classification with Caltech-101

Github: https://github.com/yeojunyoon916/SHBT_project1.git

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1. Introduction

This project evaluates image classification performance on the Caltech-101 dataset, which consists of 101 object categories and approximately 9,000 images. The primary goal is to compare classical machine learning methods against deep learning architectures, including ResNet and Vision Transformer (ViT). We investigate how architectural choices and hyperparameter configurations affect classification accuracy in a dataset characterized by significant class imbalance.

2. Methods

2.1. Dataset and Preprocessing

The dataset was split into training (70%), validation (15%), and testing (15%) sets using a stratified approach to maintain class proportions. Images were resized to 128x128 or 64x64 pixels and normalized using ImageNet statistics.

2.2. Models and Metrics

Three distinct methods were implemented:

- Classical ML: SVM using Histogram of Oriented Gradients (HOG) features extracted from 64x64 images.
- Deep Learning (CNN): ResNet-18/50 served as the baseline architecture.
- Deep Learning (Transformer): Vision Transformer (ViT) was utilized to assess attention-based feature extraction.

Evaluation metrics include overall Accuracy, Macro/Weighted Precision, Recall, and F1-score to analyze performance across imbalanced classes.

2.3. Ablation Studies

To evaluate model robustness, two ablation experiments were performed on the ResNet baseline:

1. Ablation 1: Comparison between Adam and Stochastic Gradient Descent (SGD).
2. Ablation 2: Comparison between 128x128 and 64x64 pixel input resolutions.

3. Result

The experimental results demonstrate that ResNet Ablation 1 (128x128, SGD) achieved the highest performance, reaching 0.94 Accuracy and a 0.92 Macro F1-score. This reflects a significant 21% improvement over the Adam-based ResNet baseline, which recorded 0.73 Accuracy. The classical HOG+SVM method produced the lowest results with 0.60 Accuracy.

In the ablation studies, reducing the image resolution to 64x64 (Ablation 2) caused accuracy to drop from 0.73 to 0.71, suggesting that fine-grained spatial features are necessary for distinguishing between the 101 categories. While the Vision Transformer (ViT) performed well at 0.90 Accuracy, it did not surpass the optimized ResNet model. These performance gaps are visually represented in the confusion matrices in Figure 1, where the best-performing model (C) displays the most distinct diagonal alignment.

		Precision	Recall	F1-score
SVM + HOG	Accuracy			0.60
	Macro avg	0.49	0.43	0.44
	Weighted avg	0.60	0.60	0.59
ResNET Baseline 128*128, Adam	Accuracy			0.73
	Macro avg	0.71	0.60	0.60
	Weighted avg	0.78	0.73	0.72
ResNET Ablation1 128*128, SGD	Accuracy			0.94
	Macro avg	0.93	0.91	0.92
	Weighted avg	0.95	0.94	0.94
ResNET Ablation 2 64*64, Adam	Accuracy			0.71
	Macro avg	0.66	0.59	0.60
	Weighted avg	0.74	0.71	0.71
ViT 224*224, Adam	Accuracy			0.90
	Macro avg	0.89	0.88	0.87
	Weighted avg	0.92	0.90	0.90

Table 1. Quantitative performance comparison of various classification methods on the Caltech-101 dataset.

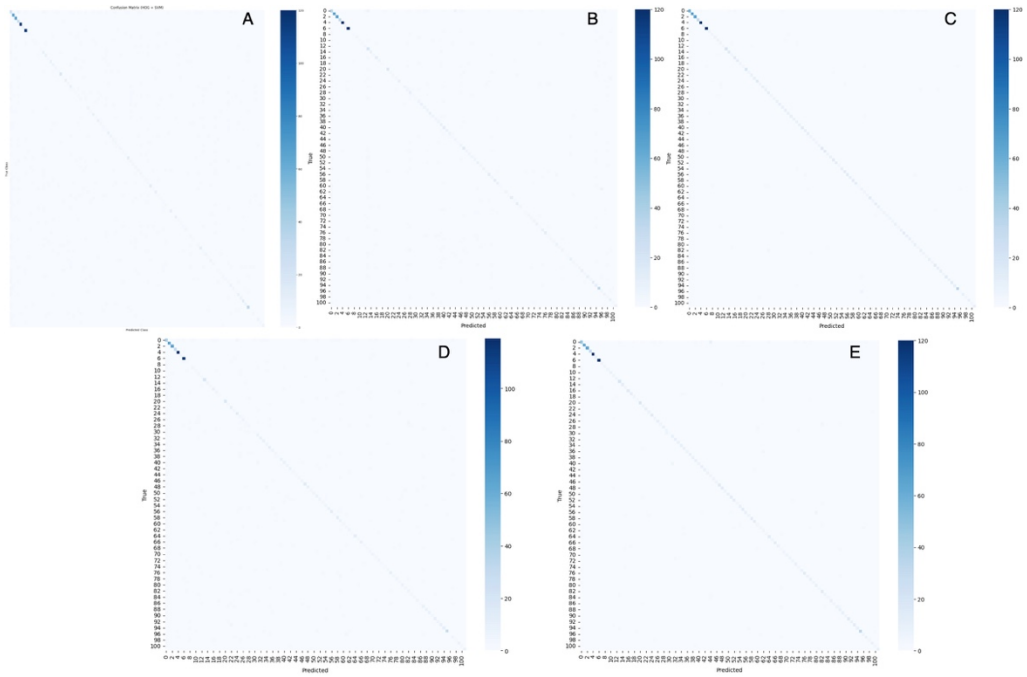

Fig 1. Confusion matrices visualizing the classification results across five different experimental setups:

(A) Classical HOG + SVM. (B) ResNET baseline - 128*128, Adam. (C) ResNET ablation 1 - 128*128, SGD. (D) ResNET ablation 2 - 64*64, Adam. (E) Vision Transformer - 224*224, Adam.

4. Dicussion

The experimental data reveals that ResNet combined with SGD (Ablation 1) achieved the highest accuracy of 0.94. This suggests that while Adam offers faster initial convergence, SGD with momentum may provide better generalization on the Caltech-101 loss surface.

Reducing input resolution to 64x64 (Ablation 2) resulted in a performance decline from 0.73 to 0.71, indicating that fine-grained spatial features are critical for distinguishing subtle category differences. Although ViT demonstrated strong performance (0.90), it remained slightly less effective than the optimized ResNet model, possibly due to the relatively small size of the dataset which may limit transformer efficiency compared to inductive biases in CNNs.

5. Conclusion

This project highlights the clear superiority of deep learning over classical feature extraction methods like HOG+SVM for complex image classification tasks. A key takeaway is that hyperparameter optimization (specifically the choice of the SGD optimizer) had a more substantial impact on final accuracy than adopting a newer architecture like ViT. Furthermore, maintainting a higher input resolution proved vital for capturing the intricate details of the Caltech-101 dataset. Future work should focus on addressing class imbalance to improve the Macro F1-scores of underrepresented labels.